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KQL as a Forensic Tool

Correlating Microsoft Logs to Reconstruct Working Time

January 2026

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About me

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Paraglider pilot

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Husband & official treat-butler for our two dogs

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Why This Topic Exists

Challanges

- Self-reported working time is subjective
- Technical logs exist but are fragmented
- Individual signals are misleading
- HR lacks technically defensible evidence

What This Solution Is NOT

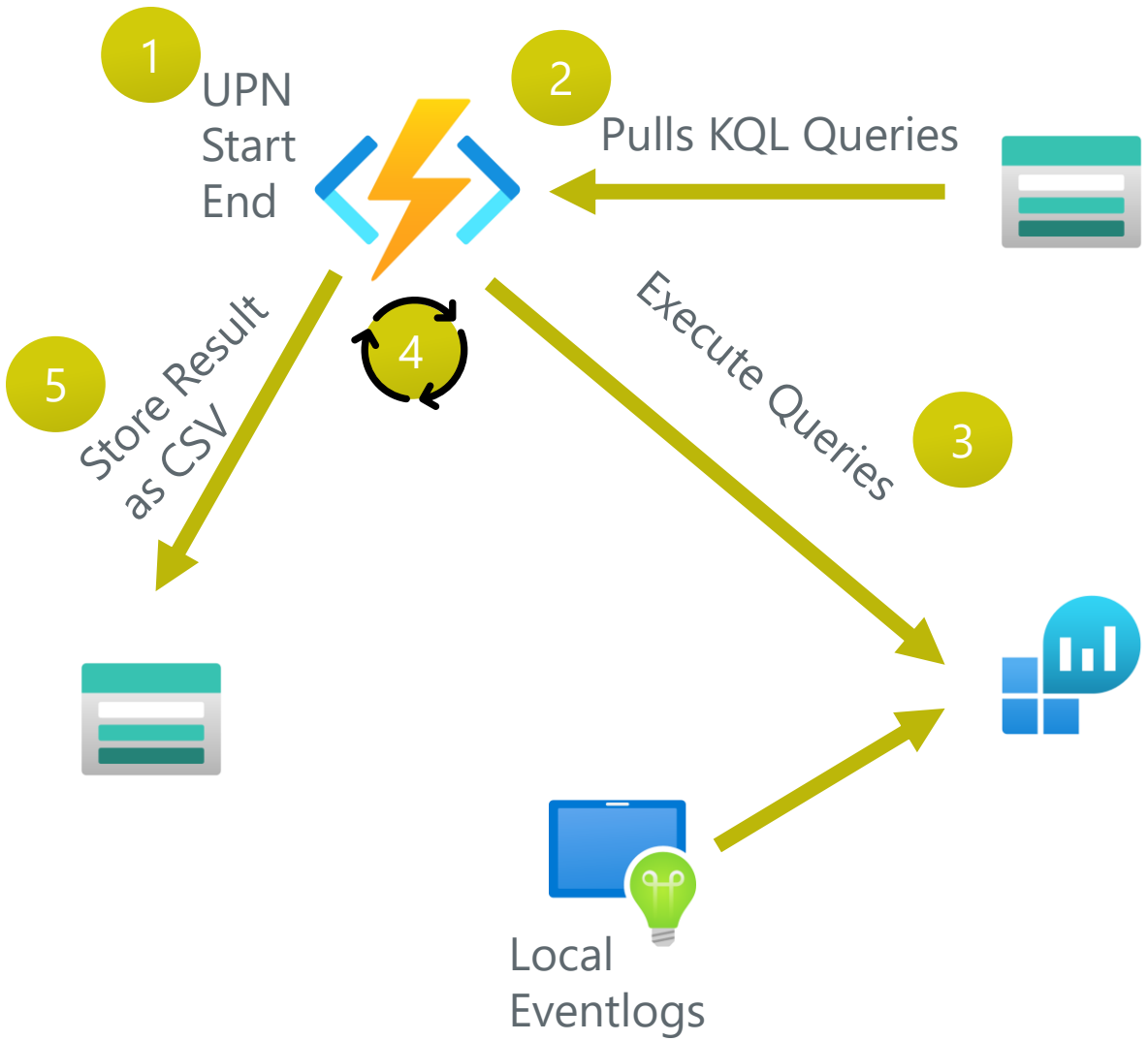
- No real-time monitoring
- No productivity scoring
- No behavioral profiling
- No automated decisions

Without Governance, This System Should Not Exist

- Purpose limitation
- Data minimization
- Strict access control
- Enforced retention limits

High-Level Architecture

- Central Log Analytics Workspace
- KQL correlation layer
- Azure Automation Runbook
- Azure Blob Storage (private, time-limited access)



Query Set

Device Timeline
Entra ID & VPN Signals
Line-of-Business App Usage
SaaS & Office Activity

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Query 1: Device Timeline

Purpose:

- Identify periods of human device interaction
- Establish potential workday boundaries

Sources:

- DeviceLogonEvents
- DeviceNetworkEvents
- DeviceEvents

Naive Device Assumptions

Common False Assumptions

- First logon = work start
- Last logoff = work end
- Device uptime = activity

Query 2: Entra ID Signals

Purpose:

- Correlate user presence across services
- Validate device activity assumptions

Sources:

- SignInLogs
- VPN / Egress data

Query 3: Local App Usage

Purpose:

- Detect active work-related processes
- Identify real interaction windows

Query 4: SaaS & Office Activity

Purpose:

- Secondary plausibility check
- Cross-validate device activity

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Why Automation Was Required

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Runbook Responsibilities

- Parameter validation
- Controlled execution
- Error handling
- Consistent output generation

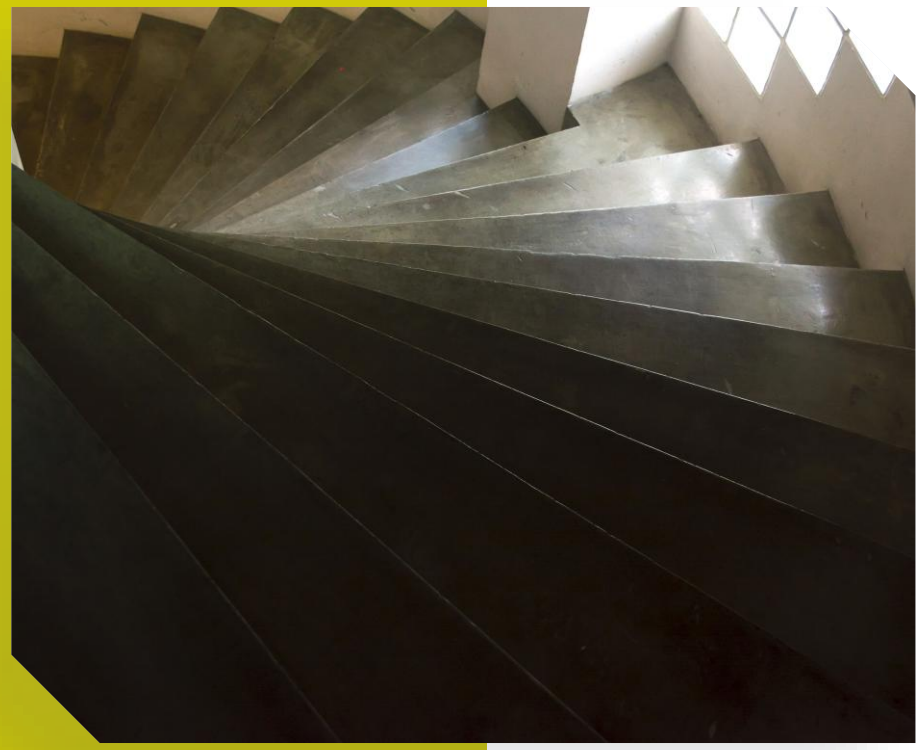
Blob Storage Strategy

Secure Storage Design

- Private containers
- SAS links with expiration
- Enforced retention policies

Deliberate Access Friction

- No emails
- No attachments
- SAS URL retrieved from runbook logs



Why Phase 2 Exists

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Scaling Governance

- Manual requests do not scale
- Traceability is required
- Consistent approval is necessary

Structured Request Flow

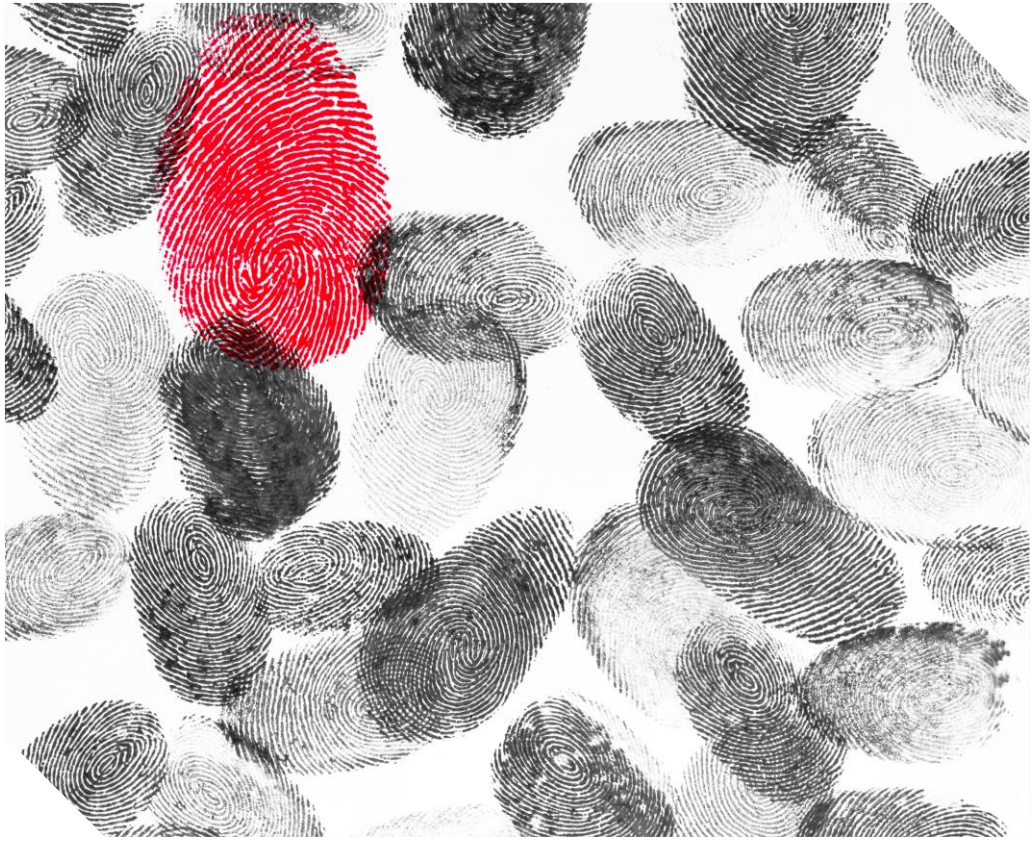
- User Principal Name
- Time range
- Validation rules
- Approval workflow

Full Traceability

- Who requested what
- When it was generated
- Where it is stored
- Who accessed it

CLOSING

What we've learned



Technical Lessons

- KQL excels at correlation
- Single-source analysis fails
- Time normalization is critical

Organizational Lessons

- HR needs interpretation guidance
- Access boundaries matter
- Documentation is non-negotiable

Technology Reveals Patterns

Humans Decide What They Mean